

Some Test Tools Available

Challenge	Tools
256 QAM and higher frequency band	➤ Agilent 89600 VSA software ➤ ZProtocol WLAN analysis software ➤ Micronics Japan Model MSA 358 8.5GHz spectrum analyser ➤ ZTEC's ZT8651 PXI/PXIe 6GHz vector signal analyser ➤ NI PXIe-5663E 6.6GHz RF vector signal analyser ➤ R&S FSW signal and spectrum analyser
Wider bandwidths	➤ R&S SMU200A and SGS100A signal generators ➤ Agilent N5182A MXG and E4438C ESG signal generators ➤ NI PXIe-5673E 6.6GHz RF vector signal generator
MIMO	➤ Scientech NB100C2 series DSO ➤ ZTEC's ZT8651 PXI/PXIe 6GHz VSA and ZProtocol WLAN ➤ Agilent 89600 VSA software, which can be used with an Infiniium or Infiniivision oscilloscope

rier leakage, phase noise in the local oscillator or amplifier non-linearity. Vector signal analyser (VSA) is a valuable tool to measure and identify the causes of poor EVM.

J.K. Baldua, director-technical, Scientech Technologies, says, "A high-resolution spectrum analyser of at least 6GHz range preferably with a tracking generator for monitoring channels, channel width, channel power, etc is required." Since 802.11ac standard utilises the 6GHz band, spectrum analysers need to be able to handle this as well as the higher bandwidth that 802.11ac requires.

Wider bandwidths

802.11ac features a wider bandwidth of 80 or even 160 MHz against the previous maximum of 40 MHz featured in the 802.11n standard. A wider bandwidth results in an improved maximum throughput for the digital communication system.

Amongst more complex challenges for design and development is the generation and analysis of wider-bandwidth signals for 802.11ac. Testing equipment able to handle 80 MHz or 160 MHz will be required to test transmitters, receivers and components.

For generating 80MHz signals, many RF signal generators do not have a high enough sampling rate to support the typical minimum 2x oversampling ratio, which can result in unnecessary images (i.e. upper sideband and lower sideband) in the signal, due to aliasing. Using proper filtering and oversampling of the waveform file, it is possible to generate 80MHz signals with good spectral characteristics and EVM.

To generate 160MHz signals, a wide-band arbitrary waveform generator (AWG), such as the Agilent 81180A, 8190A or M9330A, can be used

to create the analogue I/Q signals. These signals can then be applied to the external I/Q inputs in a vector signal generator for up-conversion to RF frequencies. It is also possible to create a 160MHz signal by using 80+80MHz mode supported by the standard to create two 80MHz segments in separate MXG or ESG signal generators and then combining the RF signals.

MIMO

MIMO is the use of multiple antennae to improve the performance of the communications system. You might have seen certain Wi-Fi routers having more than one antenna sticking out of them. These routers use MIMO technology.

Verifying MIMO design is a challenge. Multi-channel signal generation and analysis can be used to provide insight into the performance of MIMO devices and assist in troubleshooting and design verification.

For testing receivers, MIMO signals can be created with both the SystemVue WLAN library and Signal Studio. Multiple MXG or ESG signal generators can be synchronised to simulate the output of a MIMO transmitter. The effects of the fading channel can also be included in the waveform files to

provide simulation of the signals at the receiving antennae.

For MIMO transmitter test, the 89600 VSA software can be used with an Infiniium or Infiniivision oscilloscope to provide analysis of up to four channels. The analysis includes EVM and IQ measurements for all channels as well as cross-channel metrics such as the frequency response of each channel and the channel matrix. ZTEC's ZT8651 PXI/PXIe 6GHz VSA and ZProtocol WLAN offer testing covering all aspects of the 802.11ac protocol including X2, X3 and X4 MIMO streams.

Baldia says, "A digital storage oscilloscope (DSO) having 200MHz bandwidth and 1Gsps sampling is required to observe real-time signals." An example of such a DSO is the Scientech NB100C2 series.

Amplifier linearity

Amplifier linearity is the characteristic of an amplifier by which the output signal of the amplifier remains faithful to the input signal as the signal drive increases. Real amplifiers are linear only to a limit, after which the output saturates.

There are many techniques to improve amplifier linearity. Digital pre-distortion (DPD) is one such technique. Design automation software like SystemVue provide an application that simplifies and automates digital pre-distortion design for power amplifiers.

Sadaf Arif Siddiqui, technical marketing specialist from Agilent Technologies India, explains, "The SystemVue Builder software generates a stimulus waveform which is downloaded to a radio frequency (RF) signal generator and then applied to the power amplifier. The system captures the amplifier's response using a signal analyser and compares the result with the desired signal to create the pre-distortion matrix. The pre-distorted signal is then sent to the power amplifier and the response checked."

802.11ac standard devices

Products featuring the new Wi-Fi standard are starting to hit the market,